

Using Open Source and Blockchain to Build Decentralised IoT Networks

Explore how blockchain is being integrated with IoT to create decentralised networks. Find out how leading open source projects like IOTA and Streamr use blockchain to ensure data integrity, security, and privacy in IoT ecosystems.



The Internet of Things (IoT) has revolutionised how devices interact, communicate, and collect data, powering everything from smart homes and healthcare devices to autonomous vehicles and industrial automation systems. By 2030, over 50 billion devices are expected to be connected to the internet, producing an unprecedented volume of data. However, as the number of connected devices grows, the challenges associated with centralised IoT systems have become more apparent: security vulnerabilities, lack of privacy, data manipulation risks, and scalability issues.

To address these challenges, blockchain technology, coupled with decentralisation, has emerged as a promising solution. Decentralised IoT, powered by blockchain, offers

several advantages, including improved data security, privacy, and transparency, all without relying on central servers or intermediaries. Open source technologies play a crucial role in accelerating the adoption of decentralised IoT by offering flexible, transparent, and collaborative development platforms.

The basics: What is decentralised IoT?

To fully understand decentralised IoT, it's important to first break down the core concepts: IoT (Internet of Things), blockchain, and decentralisation. These are the building blocks that enable the next generation of IoT systems that are more secure, scalable, and transparent than traditional, centralised networks.